Workshop starts at 12:02

marginaleffects

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Regression Model Interpretation with marginaleffects

Instructor: David Nichols

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Logistics & Disclosure

- **Ask Questions** [in the Zoom chat].
- If you know the answer, feel free to respond (we may politely clarify if needed).
- Jillian will monitor and we will address questions after the presentation.

- **If my internet goes out:** Take a 5 minute break, and we will meet back in the same Zoom meeting.

- **Disclosure:** AI tool Claude Cowork was used in creating slides.

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From Model to Meaning

Interpreting Statistical Models
with {marginaleffects} for R

Introductory Workshop for University Researchers based on
Arel-Bundock, Greifer & Heiss | Journal of Statistical Software (2024) &
marginaleffects.com

marginaleffects

Estimands, Estimators, and Estimates

Three concepts that must be kept distinct in any rigorous statistical analysis.

Estimand

What we want to know

The quantity of scientific interest — a feature of the real world defined by the research question. Must be specified before analysis, and is independent of any statistical model or dataset.

E.g., the average effect of a financial incentive on HIV testing rates in the study population.

Estimator

How we compute it

A procedure or formula applied to data to produce a number. Two analysts can use different estimators to target the same estimand. Defined in advance, before seeing the data.

E.g., ordinary least squares (OLS), maximum likelihood.

Estimate

What we get

The specific numerical value produced when the estimator is applied to a particular dataset. Will differ across datasets drawn from the same population.

E.g., −0.039 — the estimated difference in two population means from one analysis.

> *"The main claim of this book is that the parameters of a statistical model are complex quantities that do not always have a straightforward meaning. Instead of trying to interpret them directly, we should treat parameters as a 'resting stone on the way to prediction.'"*
>
> — Arel-Bundock, Model to Meaning (2026)

A Framework for Model Interpretation: Five Questions

Instead of interpreting model parameters directly, every analysis begins by defining the quantity of interest. This requires answering five questions — organized in two steps.

Step 1 — Define the estimand (three critical questions)

1	**Quantity**	Do we want to estimate the level of a variable, the association between two (or more) variables, or the effect of a cause?
2	**Predictors**	What predictor values are we interested in?
3	**Aggregation**	Do we care about unit-level or aggregated estimates?

Step 2 — Draw conclusions (two further questions)

4	**Uncertainty**	How do we quantify uncertainty about our estimates?
5	**Test**	Which hypothesis or equivalence tests are relevant?

The Interpretation Problem

What a logistic regression returns:

Predictor	Log-odds coef.	Odds Ratio	p-value
Income ($1k)	0.32	1.38	0.004
Education (yrs)	0.85	2.34	< 0.001
Age	-0.02	0.98	0.320
Female	0.44	1.55	0.031

Research question: How much does the probability of outcome change when Education increases by one year, for a typical respondent?

The coefficient 0.85 doesn't answer this question directly.

The coefficient is on the log-odds scale. The actual probability change depends on all predictors and differs across individuals.

Why this is hard in general:

1. In nonlinear models (logit, Poisson, etc.) the effect of X is not constant — it depends on the values of all other predictors.

2. Interactions make individual coefficients uninterpretable on their own; the effect of X_1 depends on X_2.

3. Comparing effects across models or subgroups requires computing new quantities not directly output by the model.

4. Choosing where to evaluate an effect (at the mean? averaged over the data?) is a non-trivial analytical decision.

The {marginaleffects} package provides a unified set of functions to compute interpretable quantities from any fitted model, on the natural outcome scale.

The {marginaleffects} Package

What the package does

marginaleffects provides functions to compute predictions, comparisons (contrasts), slopes (marginal effects), and hypothesis tests for a wide range of fitted models. Results are returned as tidy data frames that work directly with ggplot2, dplyr, and modelsummary.

Published in the Journal of Statistical Software (2024) by Vincent Arel-Bundock, Noah Greifer, and Andrew Heiss. Available on CRAN. Supports over 100 model classes in R, with parallel support in Python.

```
citation("marginaleffects")
```

Run this in R to get the BibTeX entry.

Four core functions

`predictions()`

Predicted outcome at specified covariate values

`comparisons()`

Difference in predicted outcomes between two conditions

`slopes()`

Partial derivative of predicted outcome w.r.t. a predictor

`hypotheses()`

Test linear or nonlinear combinations of parameters

The Two Meanings of "Marginal Effect"

The term is used in two opposite senses by different research communities — a source of genuine confusion in the literature.

In economics & econometrics

Marginal = derivative

The effect of an infinitesimally small increase in X on the outcome. The slope of the prediction curve at a given point.

A differentiation concept

Formally: $\partial E[Y]/\partial X$ — the partial derivative of the regression equation. "Marginal cost" and "marginal utility" use this sense.

Conditional on X

The value depends on where you evaluate it. Different values of X give different marginal effects in nonlinear models.

`Computed by: slopes() and avg_slopes()`

≠

In statistics & biostatistics

Marginal = integrated out

An effect that has been averaged (marginalized) over the distribution of other covariates. Opposite of "conditional."

An integration concept

A "marginal model" or "marginal probability" averages over random effects, clustering, or other nuisance variables.

Population-averaged

The effect for a generic person in the population, having integrated out all individual-level variation.

`Also computed by: avg_slopes()`

avg_slopes() satisfies both definitions simultaneously: it computes derivatives (economics sense) and then averages them over the observed data distribution (statistics sense).

Installation & First Example

Install from CRAN, then load:

```
install.packages("marginaleffects")
library(marginaleffects)
mtcars$cyl <- as.factor(mtcars$cyl)
```

A minimal example — logistic regression on the built-in mtcars dataset:

```
# P(am=1): manual vs automatic
mod <- glm(am ~ hp + wt,
           data = mtcars,
           family = binomial)

# Average Marginal Effect
avg_slopes(mod)
```

```
Term Estimate Std. Error     z Pr(>|z|)    S     2.5 %   97.5 %
  hp  0.00161   0.000572  2.82  0.00483  7.7  0.000491  0.00273
  wt -0.35960   0.057147 -6.29  < 0.001 31.6 -0.471602 -0.24759
```

Interpreting the output columns:

Estimate	Std. Error	z	Pr(>|z|)	S	2.5%/97.5%
avg marginal effect on probability scale	delta-method standard error	test statistic (estimate / SE)	p-value for H0: AME = 0	surprisal (bits of evidence vs H0)	lower & upper 95% CI bounds

Interpretation: on average, 1 extra horsepower raises the probability of manual transmission by 0.16 percentage points; 1,000 extra pounds lowers it by 36 percentage points.

The Five Questions — A Framework for Every Analysis

Every marginaleffects analysis requires five decisions. The next five slides address each in turn.

1	**Quantity**	**What type of estimate?**	`predictions() / comparisons() / slopes()`
2	**Predictors**	**At which covariate values?**	`newdata = | datagrid()`
3	**Aggregation**	**Unit-level or averaged?**	`avg_*() | by = "group"`
4	**Uncertainty**	**Which standard error method?**	`vcov = | inferences(method=)`
5	**Test**	**What hypothesis to test?**	`hypothesis = | equivalence =`

Q1: Quantity — Which Type of Estimate?

The first decision determines which function to call. The three main quantities map to three functions.

`predictions()` **What outcome do we expect at given covariate values?**

$E[Y \mid X = x]$ — the conditional expectation of Y. A predicted probability for logit, a predicted mean for linear regression, a predicted count for Poisson.

Use when: you want the level of the outcome, not a change.

`comparisons()` **How does the outcome change between two conditions?**

$E[Y|X=x_1] - E[Y|X=x_0]$ — the difference (or ratio) of predictions. Supports risk differences, risk ratios, odds ratios, and log-ratios via the comparison= argument.

Use when: X is categorical, binary (treatment vs. control), or you want a finite discrete change in a continuous predictor.

`slopes()` **What is the instantaneous rate of change of the outcome?**

$\partial E[Y]/\partial X$ — the partial derivative of the regression function w.r.t. a continuous predictor. Approximates the effect of an infinitesimally small change in X.

Use when: X is continuous and you want the local rate of change (often called the 'marginal effect' in economics).

Each function also has an avg_*() variant (avg_predictions, avg_comparisons, avg_slopes) that averages unit-level estimates across the dataset.

Q2: Predictors — At Which Covariate Values?

Predictions, comparisons, and slopes are conditional quantities: their value depends on the covariate values at which they are evaluated. This choice is specified through the `newdata=` argument.

Empirical grid (default)

One estimate per row of training data. Averaging gives the AME, which marginalizes over the observed predictor distribution.

Mean / mode

All predictors fixed at their mean or mode. Gives the marginal effect at the mean (MEM). The 'mean person' can be unrealistic.

User-specified grid

Fully custom profiles; unspecified predictors default to mean or mode. Useful for targeted 'what if' scenarios.

datagrid() examples:

```
mod <- glm(am ~ hp + wt + cyl,
           data = mtcars,
           family = binomial)

# Specific hp values; wt at its mean
avg_slopes(mod, variables = "hp",
  newdata = datagrid(hp = c(100, 200)))

# Cross product: 2 hp × 2 wt values
avg_slopes(mod, variables = "hp",
  newdata = datagrid(hp = c(100, 200),
                     wt = c(2.5, 4.0)))

# Marginal effect at the mean
avg_slopes(mod, newdata = "mean")
```

The choice of grid is a substantive decision, not just a technical one — different grids estimate different quantities and can yield numerically different results.

Q3: Aggregation — Unit-Level or Averaged?

Every observation has its own slope or comparison. Whether to report all of them or summarize them is the aggregation decision.

Unit-level:

Returns one estimate per observation. Useful for diagnostics, but too many rows for a paper.

Sample average:

Average of all unit-level estimates. The AME — a single statistic reflecting the observed predictor distribution.

Average by subgroup:

Computes the AME for each group level. Reveals heterogeneous effects; use hypothesis= to test differences.

Weighted average:

Weights estimates before averaging. Useful for sampling weights or reweighting to a target population.

Code examples:

```
mod <- glm(am ~ hp + cyl, data = mtcars,
  family = binomial(link = "probit"))

# Unit-level (one per observation)
slopes(mod, variables = "hp")

# Sample average (the AME)
avg_slopes(mod, variables = "hp")

# Average by subgroup
avg_slopes(mod, variables = "hp",
           by = "cyl")

# Formal equality test
avg_slopes(mod, variables = "hp",
  by = "cyl", hypothesis = "b2 = b3")
```

In most research contexts, report the Average Marginal Effect (avg_slopes). Use by= to explore and test heterogeneity.

Q4: Uncertainty — Which Standard Error Method?

Standard errors for marginal effects are not directly available from the model's coefficient covariance matrix — they require propagation through the prediction function.

Delta method (default)

Propagates uncertainty from the model's covariance matrix via first-order Taylor expansion. Fast; adequate for large samples.

Robust / heteroskedasticity-consistent

Uses a sandwich covariance estimator. Appropriate when the variance-covariance assumption of the model is violated.

Clustered standard errors

Clusters SEs at the level of a grouping variable. Required when observations within clusters are not independent.

Bootstrap / simulation

Resamples data to construct confidence intervals. More reliable for small samples or when the delta method is a poor approximation.

Code examples:

```
mod <- glm(am ~ hp + wt,
  data = mtcars, family = binomial)

# Default (delta method)
avg_slopes(mod)

# HC3 robust SEs
avg_slopes(mod, vcov = "HC3")

# Clustered SEs
avg_slopes(mod, vcov = ~carb)

# Bootstrap (500 resamples)
avg_slopes(mod) |>
  inferences(method = "boot", R = 500)
```

The same estimate with different vcov choices produces different confidence intervals. The choice should be driven by the data-generating process, not convenience.

Q5: Test — What Hypothesis to Test?

marginaleffects supports null hypothesis tests and equivalence tests on any estimate or function of estimates, using the `hypothesis=` and `equivalence=` arguments.

Test against zero (default)

Each estimate is automatically tested against H_0: estimate = 0. This is the default output of every marginaleffects function.

Test against a non-zero value

`hypothesis = "b1 = 0.01"` tests whether estimate 1 differs from 0.01 — useful against a minimum detectable effect or published benchmark.

Test equality of two estimates

`hypothesis = "b1 = b2"` or `"b1 - b2 = 0"` tests whether two row estimates (e.g., AMEs for two subgroups) are significantly different.

Equivalence test (TOST)

`equivalence = c(-d, d)` tests whether an estimate lies within [−d, d] using the two one-sided test (TOST) procedure.

Code examples:

```
mod <- glm(am ~ hp + wt,
  data = mtcars, family = binomial)

# Equal AMEs for hp and wt?
avg_slopes(mod, hypothesis = "hp = wt")
# Test vs. benchmark (0.003)
avg_slopes(mod, variables = "hp",
  hypothesis = "b1 = 0.003")

# Are subgroup AMEs equal?
avg_slopes(mod, variables = "hp",
  by = "cyl", hypothesis = "b1 = b2")

# Equivalence: within ±0.002?
avg_slopes(mod, variables = "hp",
  equivalence = c(-0.002, 0.002))
```

The hypothesis= argument accepts any equation-like string. b1, b2, ... refer to rows of the output in order. Term names (hp, wt) can be used by name.

AME vs. MEM — A Critical Statistical Distinction

Two summaries of a slope — they answer subtly different questions and can give different numbers.

Marginal Effect at the Mean (MEM)

Evaluates the slope at the point where every predictor equals its mean (or mode for categorical variables).

Formula: $g'(\beta_0 + \beta_1\bar{x}_1 + \beta_2\bar{x}_2 + \ldots) \cdot \beta_j$

The 'mean person' is a mathematical construct that may not correspond to any real individual in the data.

In Jensen's inequality terms: $g(E[X]) \neq E[g(X)]$ — in nonlinear models, the slope at the mean is generally not equal to the mean slope.

In marginaleffects:
slopes(mod, newdata = "mean")

VS

Average Marginal Effect (AME)

Computes the slope for every individual in the dataset, then averages those slopes.

Formula: $(1/N)\ \Sigma_i\ g'(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \ldots) \cdot \beta_j$

Averages over the actual empirical distribution of covariates. More robust to Jensen's inequality. Preferred when the dataset is representative of the target population.

In marginaleffects:
avg_slopes(mod)

Recommendation: prefer the AME (avg_slopes) in most published research. Report the MEM only when you have a theoretical reason to evaluate effects at the mean covariate profile.

predictions() and comparisons() in Practice

Both functions use the same argument structure. The key difference is what question they answer.

predictions()

```r
mod <- glm(am ~ hp + wt, data = mtcars,
           family = binomial)

# Predict at specific covariate values
predictions(mod,
  newdata = datagrid(hp = c(100, 200),
                     wt = 3.0))

# Average predicted probability
avg_predictions(mod)
```

Reading predictions() output

estimate is E[Y|X=x]: a predicted probability, mean, or count on the response scale. CIs are in the same units.

comparisons()

```r
# Default: approx. +1 unit change per predictor
avg_comparisons(mod)

# Discrete change: wt from 2.5 to 3.5
avg_comparisons(mod,
  variables = list(wt = c(2.5, 3.5)))

# Ratio instead of difference
avg_comparisons(mod,
  variables = "wt",
  comparison = "ratio")
```

Reading comparisons() output

estimate is $E[Y|X=x_1] - E[Y|X=x_0]$: the change in predicted outcomes. For logistic or probit models this is a probability difference.

Use predictions() when you need levels; use comparisons() when you need changes. comparisons() subsumes the classical 'average treatment effect on the treated' estimand.

slopes() in Practice — Marginal Effects

slopes() computes the partial derivative ∂E[Y]/∂X. The key insight is that in nonlinear models this derivative is not constant — it changes with the covariate values.

Common usage patterns:

```
mod <- glm(am ~ hp + wt, data = mtcars,
           family = binomial)

# Unit-level slopes
slopes(mod, variables = "hp")

# Average Marginal Effect (AME)
avg_slopes(mod, variables = "hp")

# AME by subgroup
avg_slopes(mod, variables = "hp",
           by = "cyl")
```

Interpretation notes:

The slope is a local quantity

∂E[Y]/∂X is the slope at a specific point. In a logistic model it is steepest at P=0.5 and flat near P=0 or P=1.

AME approximates a one-unit change

avg_slopes() can be loosely interpreted as the average change in E[Y] for a one-unit increase in X — valid only for small changes.

Slopes for categorical predictors

slopes() switches to comparisons() semantics when the focal variable is categorical or a factor. Output is then a contrast, not a derivative.

Slopes are conditional: always report where they were evaluated. plot_slopes(condition='X') is often more informative than a single average.

Visualization — plot_predictions() / plot_slopes()

All plot_* functions return ggplot2 objects — customize freely with geom_*, theme_*, labs(), and patchwork.

```
library(marginaleffects)
library(ggplot2)
install.packages("patchwork")   # run once to install
library(patchwork)    # provides the / operator

mod <- glm(am ~ hp + wt, data = mtcars, family = binomial)

# Prediction curve
p1 <- plot_predictions(mod, condition = "hp")

# Slope of hp across its range
p2 <- plot_slopes(mod, variables = "hp", condition = "hp")

# Stack vertically (patchwork syntax)
p1 / p2
```

Three visualization functions:

`plot_predictions()`

Prediction curve with confidence ribbon. Shows level of outcome over a covariate range.

`plot_comparisons()`

Contrast or treatment effect over a predictor range or by group.

`plot_slopes()`

Marginal effect plotted as a function of a predictor. Reveals non-linearity.

The condition= argument sets the x-axis variable and controls color/facet groupings.

Visualize before averaging: plot_slopes(condition='X') often reveals non-linearity that a single AME hides.

The Five Questions Applied — A Complete Workflow

Research question: Does distance to an HIV test center reduce the probability of seeking test results, and does a financial incentive moderate this effect?

```
library(marginaleffects)
dat <- get_dataset("thornton")
mod <- glm(outcome ~ incentive * distance,
           data = dat, family = binomial)

# Q1 Quantity:    slope w.r.t. distance
# Q2 Predictors:  empirical grid (default)
# Q3 Aggregation: average by incentive
# Q4 Uncertainty: cluster by village
# Q5 Test:        are group slopes equal?

avg_slopes(mod,
  variables  = "distance",      # Q1
  newdata    = NULL,            # Q2
  by         = "incentive",     # Q3
  vcov       = ~ village,       # Q4
  hypothesis = "b1 - b2 = 0")   # Q5
```

Partial Output:

```
 Hypothesis Estimate     SE      z p-value
    b1-b2=0  -0.0142 0.0165 -0.861    0.39
```

Interpretation:

Q1 + Q3 The average slope of distance w.r.t. outcome is computed separately for each incentive group.

Q4 Standard errors are clustered at the village level, accounting for within-village correlation.

Q5 The difference is −0.0142 (SE = 0.0165, z = −0.861, p = 0.39). We cannot reject that distance has the same effect in both groups.

Each argument in the call directly addresses one of the five questions. Reading the code and the framework together tells the full analytic story.

Supported Model Types — Over 100 Classes

marginaleffects works with any model class for which a predict() method exists. The same five-question framework applies regardless of the modeling package.

Linear models

lm, rlm, glm
Gaussian / Poisson / Gamma
Negative binomial, quasipoisson
betareg

Mixed effects

lme4: lmer, glmer
nlme: lme, gls
glmmTMB, lmerTest
brms (Bayesian)

Survival

survival: coxph, survreg
flexsurv: flexsurvreg
Parametric & Cox models

Flexible / smooth

mgcv: gam, bam
gamlss
nnet: multinom
ordered: polr, clm

Causal / matching

MatchIt + glm
IPW estimators
G-computation workflows
(see book Ch. 8–9)

Machine learning

bart
tidymodels / parsnip
caret
Scikit-learn (Python)

`Full list: marginaleffects.com/bonus/supported_models.html`

Resources & Key Takeaways

Five takeaways:

1. Coefficients are parameters, not effects. In nonlinear models they do not describe how the outcome changes.
2. Every analysis requires five decisions: Quantity, Predictors, Aggregation, Uncertainty, and Test.
3. avg_slopes() and avg_comparisons() express effects on the natural outcome scale (probabilities, means, counts).
4. The AME averages over the observed covariate distribution; the MEM evaluates at the mean. They are not equivalent.
5. The hypothesis= argument lets you test any estimand — group differences, custom contrasts, equivalence bounds.

Resources:

Web book (free)
`marginaleffects.com/chapters/who.html`
Full text of Model to Meaning by Arel-Bundock (2026, CRC Press)

Quick start
`marginaleffects.com/bonus/get_started.html`
A 30-minute introduction covering all four main functions

JSS article
`doi.org/10.18637/jss.v111.i09`
Open-access paper describing the conceptual framework formally

Case studies (R)
`marginaleffects.com/bonus/logit.html   (and more)`
Logit, mixed effects, GAM, survival, experiments, G-computation...

Exercise 1 — Logistic Regression Interpretation

mtcars dataset (built into base R)

Starter code:

```
library(marginaleffects)
mtcars$cyl <- as.factor(mtcars$cyl)


mod <- glm(am ~ hp + wt + cyl,
           data = mtcars,
           family = binomial)

# Inspect the model
summary(mod)
```

Discussion:

The hp coefficient in summary(mod) is on the log-odds scale. The AME from avg_slopes() is on the probability scale. Which is more useful to report?

Questions:

Q1 What is the predicted P(am=1) for a 4-cylinder car with hp=100 and wt=2.5?

```
predictions(mod, newdata = datagrid(
  cyl=4, hp=100, wt=2.5))
```

Q2 What is the Average Marginal Effect of hp? Write one sentence interpreting it.

```
avg_slopes(mod, variables = "hp")
```

Q3 How does the slope of wt change as wt itself increases? (Hint: visualize it.)

```
plot_slopes(mod, variables = "wt",
            condition = "wt")
```

Exercise 2 — Comparisons, Subgroups & Hypothesis Tests

Continue with the model from Exercise 1 | 10 minutes

Run this code:

```
mod <- glm(am ~ hp + wt + cyl,
           data = mtcars,
           family = binomial)

# Q1: discrete change in wt
avg_comparisons(mod,
  variables = list(wt = c(2, 2.5)))

# Q2: AME of hp by cylinder
avg_slopes(mod,
  variables = "hp",
  by = "cyl")

# Q3: formal test of equality
avg_slopes(mod,
  variables = "hp",
  by = "cyl",
  hypothesis = "b1 = b3")
```

Questions:

Q1 When wt goes from 2 to 2.5, what happens to P(am=1)? Interpret the estimate and its confidence interval.

Q2 Is the AME of hp larger for 4-cylinder or 8-cylinder cars? What might explain this pattern?

Q3 Is the difference between the AME for cyl=4 and cyl=8 statistically significant? What does the p-value tell you?

Reflection: Which of the five framework questions did each of these three code chunks address?